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Subject :- Computer Network Lab

1) What are the different protocols you observe at the following layers of protocol stack?

After following the given instruction in the document I am able to capture 16700 packets within 6 mins.

i) Application layer

In 16700 packets, found 497 (3.0%) of application protocols which include

- HTTP / JSON $\rightarrow$ Contain 1 packet is less than 0.1%.
- DHCP $\rightarrow$ Containing 19 packets i.e 0.1%.
- DNS $\rightarrow$ Contain 2.8% i.e 462 packets.
- HTTP $\rightarrow$ Contain 16 (0.1%) packets

- SSDP $\rightarrow$ Contain 0.3%. i.e 55 Packets.
- Link $\rightarrow$ local multicast name resolution
 $\rightarrow$ Contain 20%. i.e packet in each
 IPV4 & IPV6.
- Multicast DNS $\rightarrow$ 9.5%. i.e 430 packet in
 IPV6 & 2.5%. i.e 420 packet in IPV4.
- Dynamic host Configuration Protocol $\rightarrow$ Contain
 0.1%. i.e 19 packet in IPV4 &
 0.1%. i.e 15 packets in IPV6.

ii) Transport layer :-

- TCP $\rightarrow$ Contain 38.6%. i.e 69393 packets.
- UDP $\rightarrow$ Contain 26.2%. i.e 438 packets
 in IPV4 & 2.7%. i.e 455 packet in
 IPV6.

iii) Network layer :-

- Internet group management protocol $\rightarrow$
 Contain 0.6%. i.e 95 packet.
- Internet Control message protocol
 (ICMPv6) $\rightarrow$ Contain 1.6%. i.e 248
 packet.
- IPV4 : Contain 65%. i.e 10305 packets
- IPV6 : Contain 4.2%. i.e 703 packets

Q.2) What is the total amount of data being received for the following two cases?

1) Newer881.com → IP Add → [34. 228. 124 . 45]

Step → Filter . IP address for specific website used IP add = 34. 228. 124. 40

Step II → Statistics → Conversation & then:
50 found total 200 packet Conversion
b/w & total amount of data received
is if packet which is 32 KB.

2) Cornell.edu : IP Address → [128. 253. 173. 241]

I found total 72 Packet of 29 KB & total packet received from Cornell.edu is 22 KB of total packet 36.

Q.3) Answer the following when you access the site newer881.com →

1) What Version of HTTP the Server running -g ?

→ only HTTP /

2) How many HTTP GET request do you observe ?

→ Total 6 HTTP GET request from newer881.com

* HTTP GET request are :-

- i) 579 GET /online HTTP/1.1
- ii) 580 GET /online HTTP/1.1
- iii) 485 GET /favicon.ico HTTP/1.1
- iv) 585 GET /online HTTP/1.1
- v) 559 GET /online HTTP/1.1
- vi) 495 GET /online HTTP/1.1

∴ By using "http request . method" = "GET"
and I.P address = 34.223.124.75

- 3) Identify TCP three way handshake used to establish the connection with ACK P-
ackets & write the sequence no of
each of the three access.

Solve :- Get request

No → 3295 → GET /online HTTP/1.1
Stream no → 101

<u>Number</u>	<u>information</u>	<u>Sequence No</u>
3198	SYN	0
3295	SYN-ACK	0
3296	ACK	1

Step which I follow to answer

Step 1 → Get the details of GET HTTP request.

"http.request.method == "GET"

Step 2 → Select any request & check stream index from panel.

Step 3 → To get TCP handshake details filter using this command →

"tcp.stream == 101"

Step 4 → Get the information from info column & non-column & panel for sequence no.